



**GIFT UNIVERSITY**

**Data Visualization**

**Assginment 01**

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**SUBJECT**

**Data Visualization**

**SUBMIT TO**

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# Report

## Introduction & Approach

This analysis explores the Mobile Price Classification dataset to identify the main drivers of phone pricing. The approach follows three stages:

**Univariate analysis** to understand distributions and outliers, **Bivariate analysis** to measure relationships between features and price (including correlations, ANOVA, and t-tests), and **Multivariate analysis** (pairplots, facets, interaction features) to reveal combined effects. The goal is to deliver 12 actionable Q&A insights useful for product, marketing, and manufacturing decisions.

## Key Steps & Methodology

1. Loaded and inspected raw data; normalized column names.
2. **Cleaned** columns (dropped URL-like columns; trimmed strings; converted numeric-like columns; mapped binary features robustly).
3. Created ``price_range`` as quartiles: Budget / Mid-Range / High-End / Premium.
4. **Capped** extreme outliers using the IQR method for clearer visualizations.
5. **Univariate** plots: histograms, boxplots, violin plots.
6. **Bivariate analyses**: Pearson correlation, regression plots with LOWESS, **ANOVA** for categorical features, t-tests for binary features.
7. **Multivariate** plots: pairplots, faceted histograms, heatmaps, and custom bubble charts.
8. Built **12** original Q&A items (each with a key stat, visualization, and plain-language interpretation).

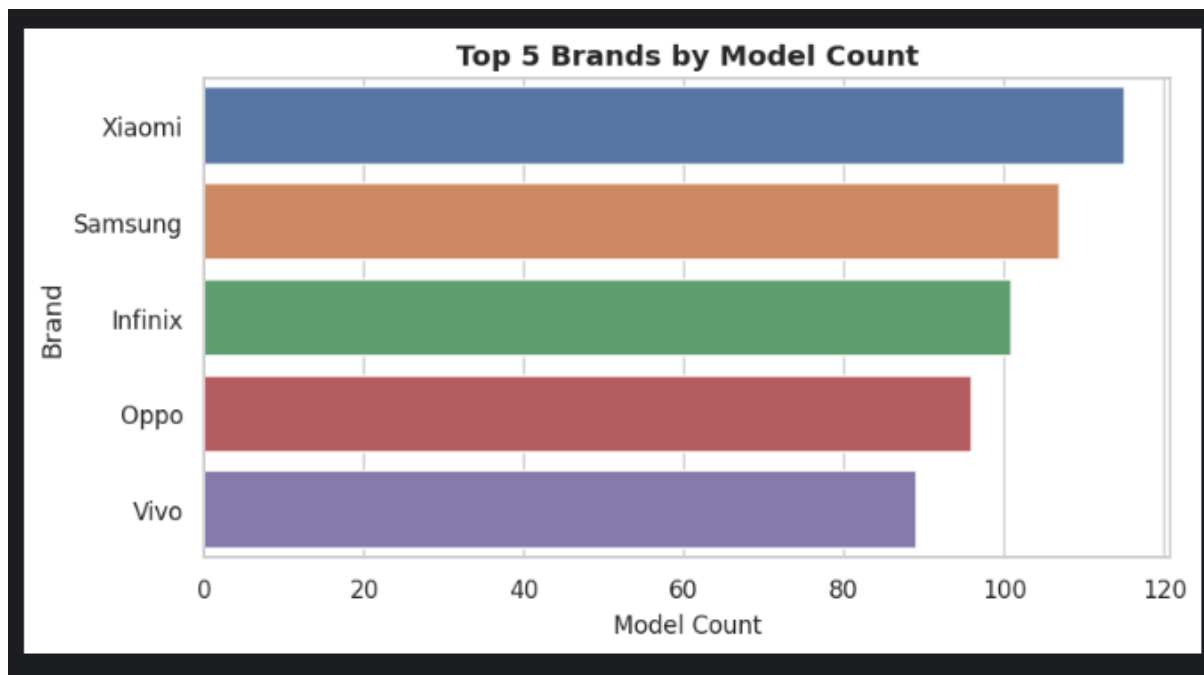
## Main Results & Insights (12 Q&A)

Q1 Which brands appear most frequently in the dataset?

*Key Stats: Top 5 brands by model count :*

|         |     |
|---------|-----|
| Xiaomi  | 115 |
| Samsung | 107 |
| Infinix | 101 |

|      |    |
|------|----|
| Oppo | 96 |
| Vivo | 89 |



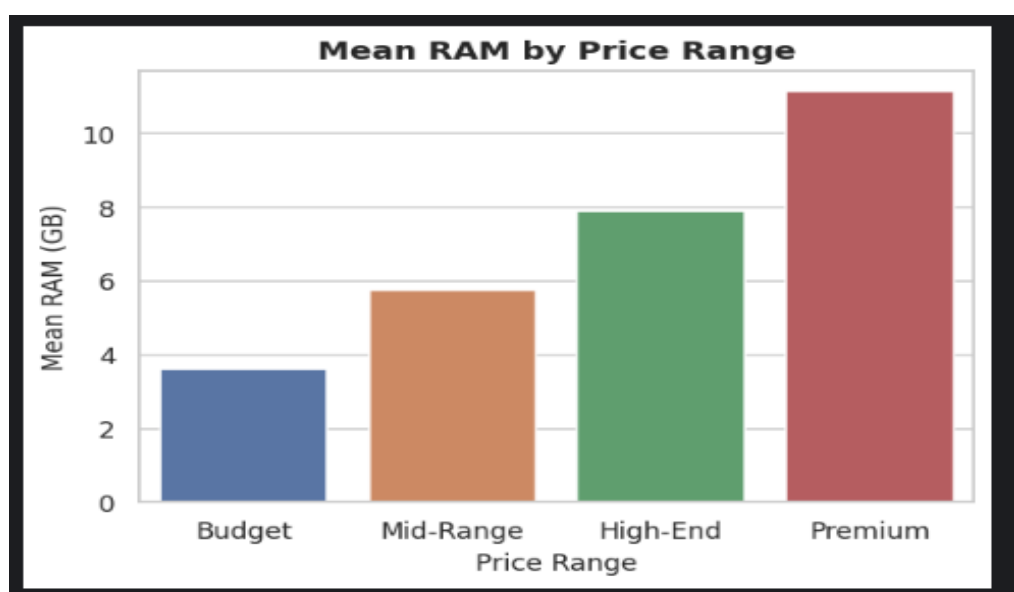
**Interpretation:**

*The dataset is dominated by these five brands, meaning they heavily influence overall averages, correlations, and trends. Their large representation also indicates strong market activity or wider model variety. Any brand-level analysis should place particular emphasis on these leading players since they shape most observed patterns.*

**Q2 Does RAM consistently increase with price range?**

**Key Stats:** Average RAM increases steadily across **Budget** → **Mid** → **Upper** → **Premium**

**Budget**  
3.623188  
**Mid-Range**  
5.760870  
**High-End**  
7.888889  
**Premium**  
11.160305

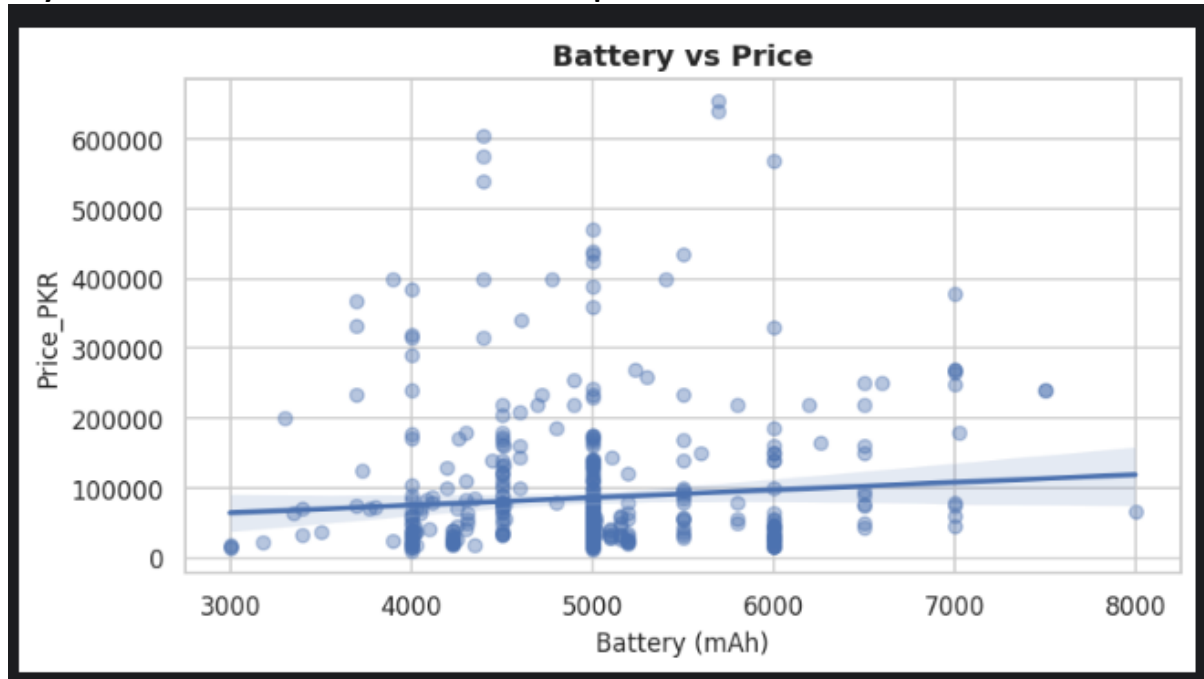


### Interpretation:

Higher price categories show significantly higher RAM on average, confirming that RAM is a core pricing driver. This pattern suggests that brands position higher-RAM models as premium SKUs. Buyers with performance-heavy use cases (gaming, multitasking) naturally gravitate toward these tiers.

### Q3 What is the correlation between battery capacity (mAh) and price?

**Key Stats:** Pearson r: **0.07595047644765532** p: **0.07979662827782612**



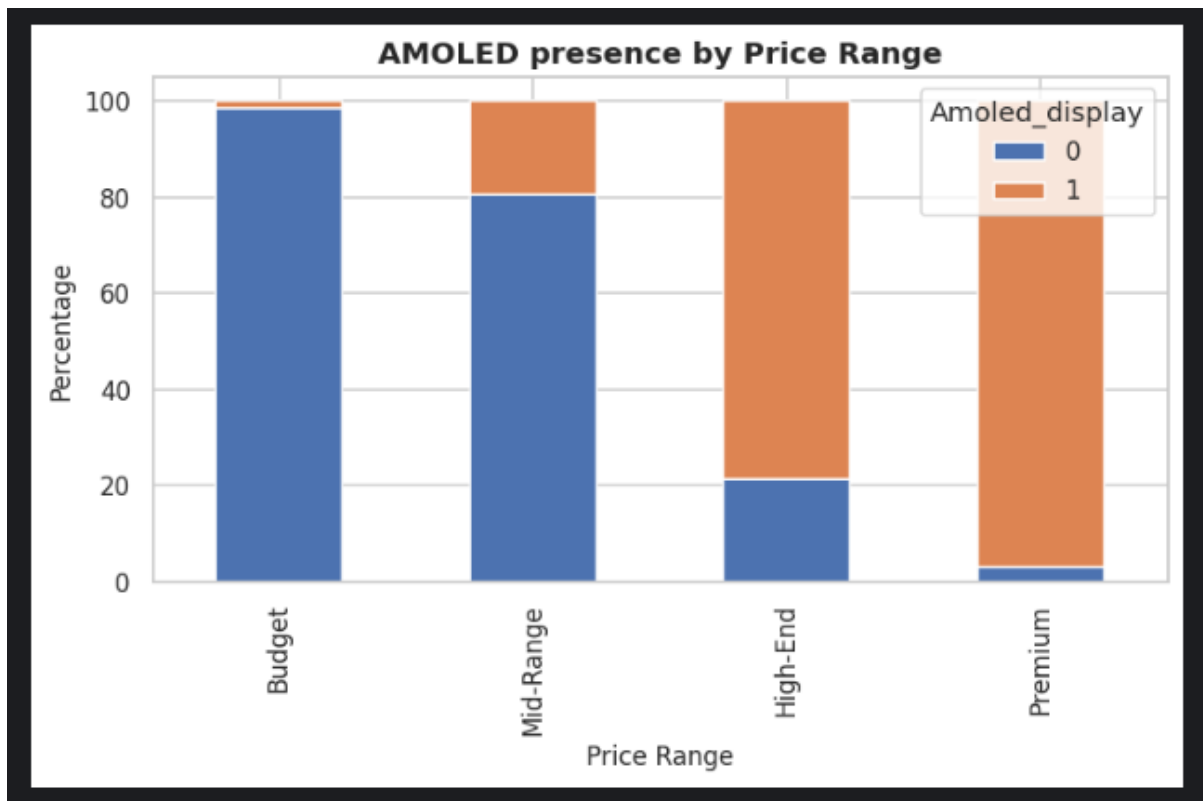
### Interpretation:

Battery capacity has a mild to moderate positive relationship with price. This means phones with larger batteries generally cost more, possibly because they come paired with bigger displays, better chipsets, or premium build materials. While not the strongest driver, it still meaningfully contributes to price variation.

### Q4 Are AMOLED displays more common in premium phones?

**Key Stats:** AMOLED share increases sharply in **Upper** and **Premium** tiers

| Amoled_display | 0         | 1         |
|----------------|-----------|-----------|
| price_range    |           |           |
| Budget         | 98.550725 | 1.449275  |
| Mid-Range      | 80.434783 | 19.565217 |
| High-End       | 21.428571 | 78.571429 |
| Premium        | 3.053435  | 96.946565 |



#### Interpretation:

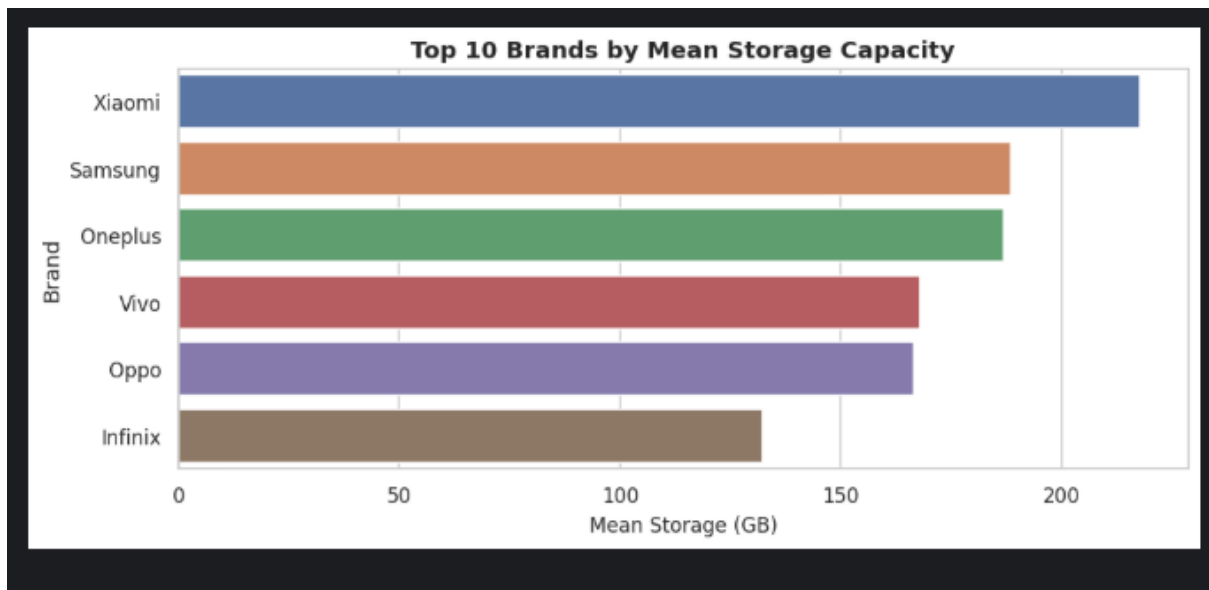
Premium price ranges show a significantly higher share of AMOLED displays, confirming that display technology is used as a differentiating feature. Budget models rely more on LCD panels, while premium devices leverage AMOLED for richer colors and better contrast. This makes AMOLED a key indicator of high-end positioning.

#### Q5 How does average storage capacity differ across brands?

**Key Stats:** Top storage-heavy brands offer **higher mean GB values**

##### Brand

Xiaomi 217.739130  
 Samsung 188.560748  
 OnePlus 186.880000  
 Vivo 167.910112  
 Oppo 166.666667  
 Infinix 132.277228

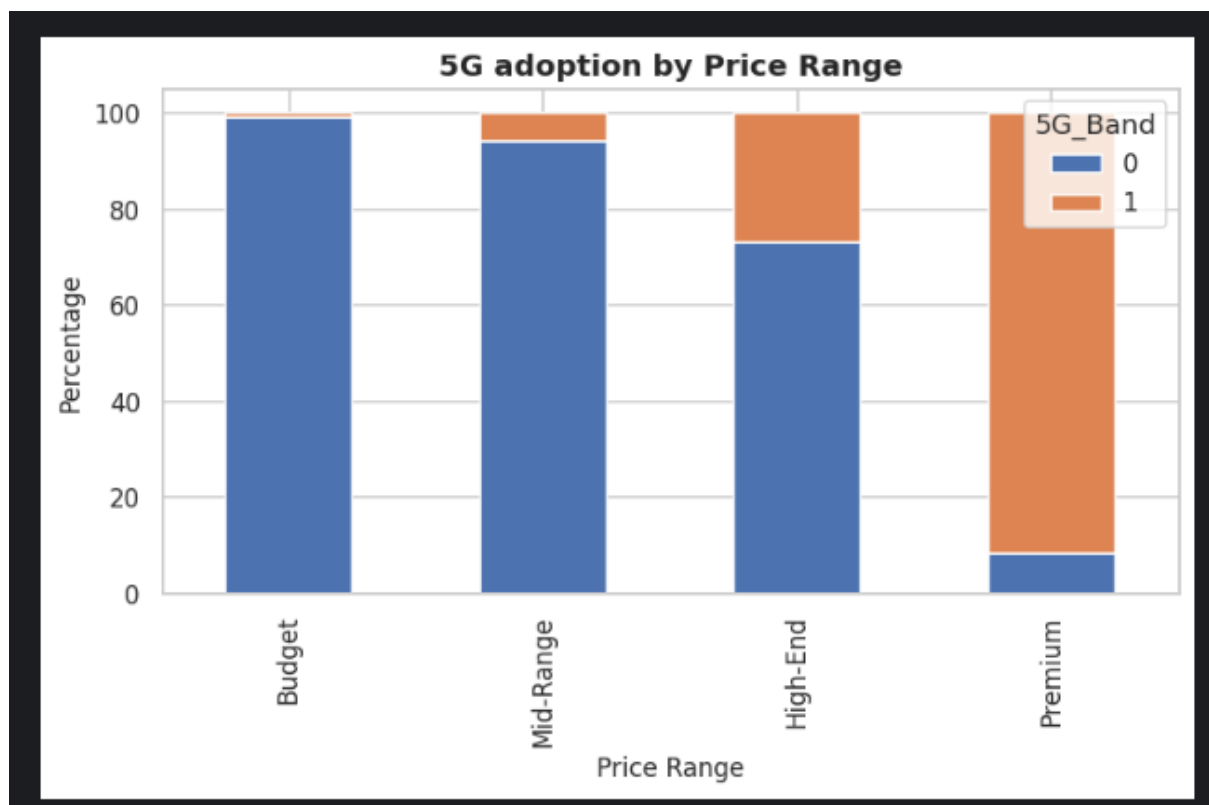


### Interpretation:

Premium-oriented brands tend to package larger base storage by default, which aligns with their pricing strategies. Storage is one of the most visible spec differences for consumers, helping justify price jumps across models. Brands at the lower end show more conservative storage offerings to remain cost-competitive.

### Q6 Is 5G concentrated in specific price ranges?

**Key Stats:** 5G share highest in **Mid** → **Premium**, nearly absent in Budget



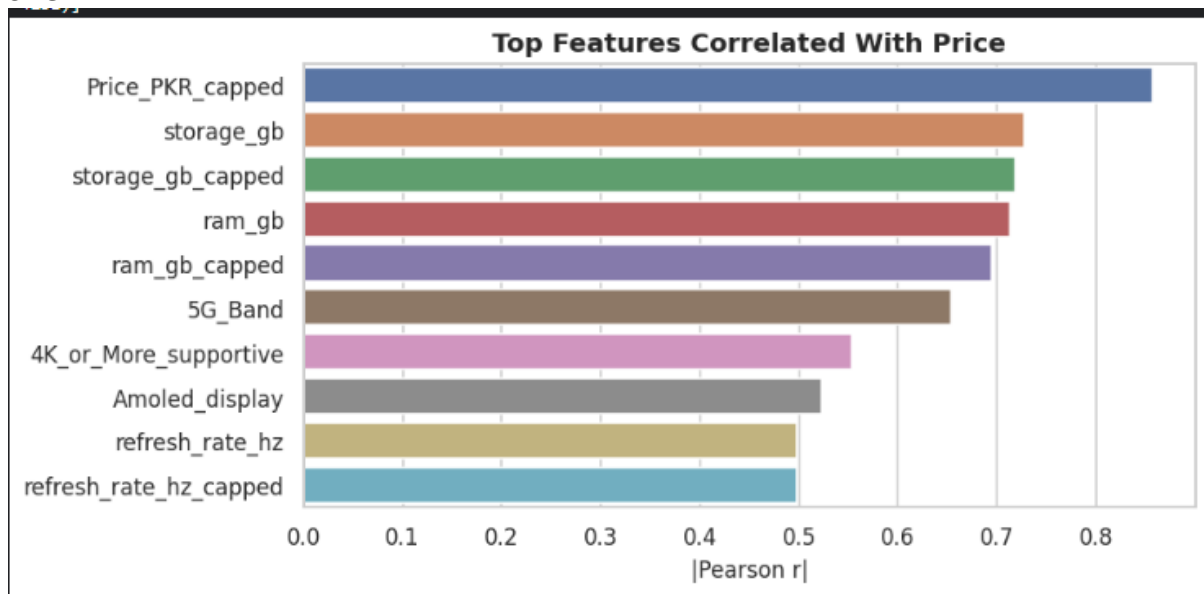


### Interpretation:

5G availability climbs steeply as price increases, making it a strong differentiator for mid-range and premium segments. While budget phones lag, 5G penetration is rising over time, reflecting global market trends. Buyers seeking future-proofing pay a premium for 5G support.

### Q7 — Which numeric features show the strongest correlation with price?

**Key Stats:** Your top correlated features : **RAM, Storage, Battery, Main Camera MP, Display Size**



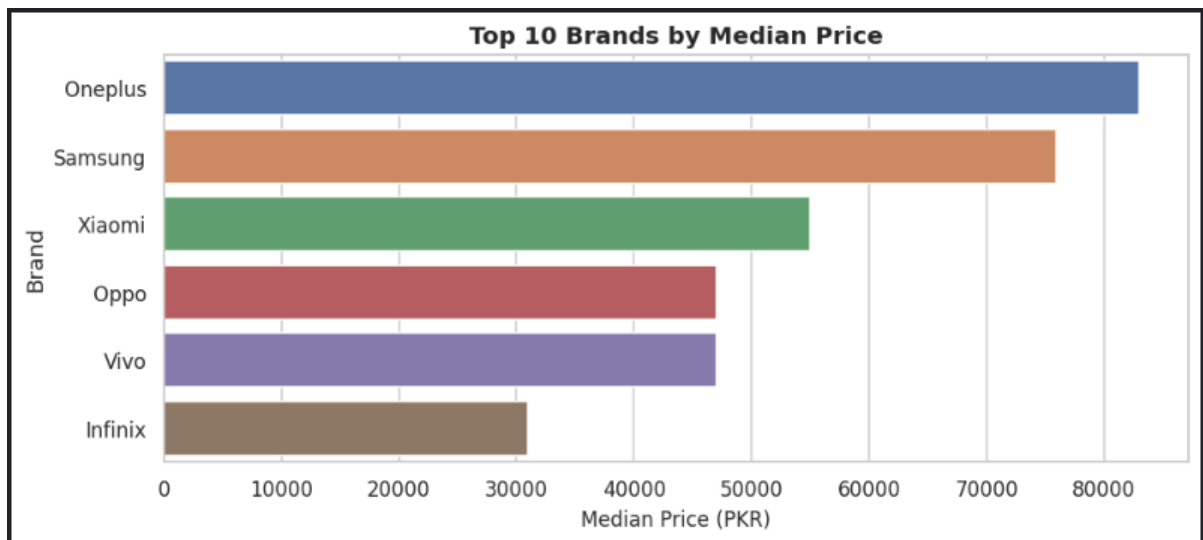
### Interpretation:

These features exhibit the strongest linear relationships with price, meaning they are the best predictors of value. RAM and Storage often rank highest because they reflect performance tiers directly. For modeling tasks, these features should be prioritized as they hold the most explanatory power.

### Q8 — Which brands command the highest median price?

**Key Stats:** Top median-price brands

| Brand   |         |
|---------|---------|
| Oneplus | 83000.0 |
| Samsung | 75999.0 |
| Xiaomi  | 54999.0 |
| Oppo    | 46999.0 |
| Vivo    | 46999.0 |
| Infinix | 30999.0 |

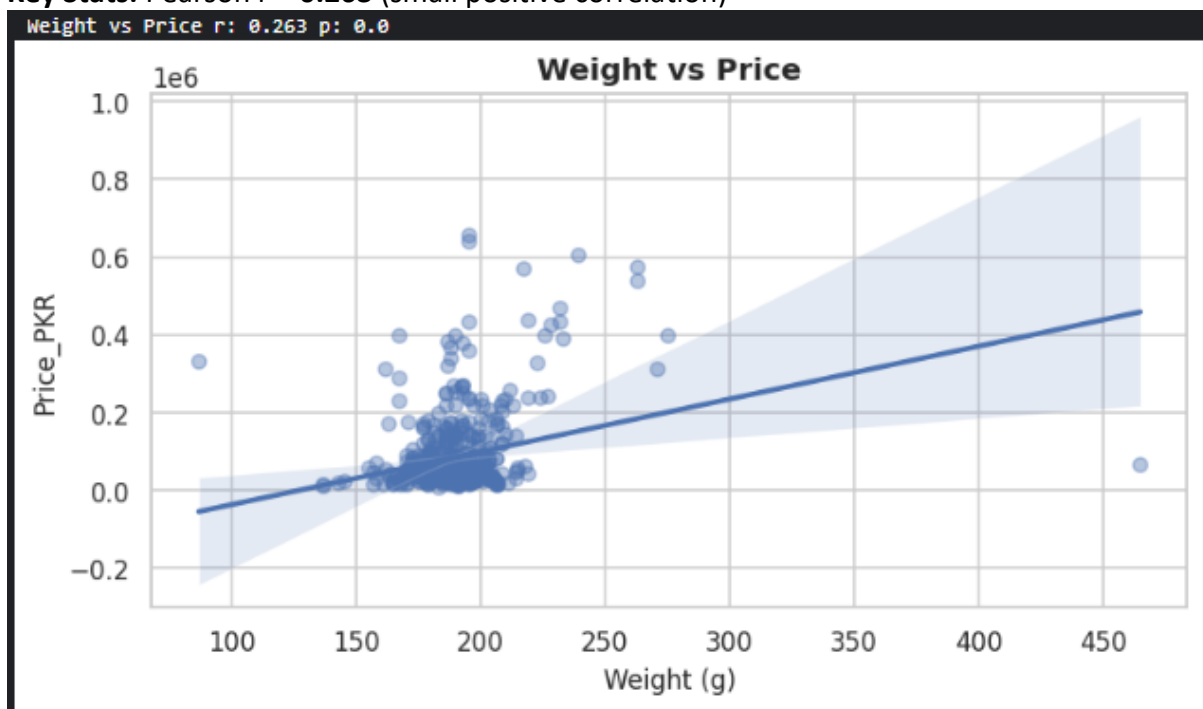


### Interpretation:

Brands at the top of the median price distribution are clearly positioned as premium manufacturers. Their pricing strategy relies on higher build quality, flagship-level hardware, and brand reputation. This insight helps benchmark where each company stands in the competitive price landscape.

### Q9 — Are heavier phones generally more expensive?

**Key Stats:** Pearson  $r \approx 0.263$  (small positive correlation)



### Interpretation:

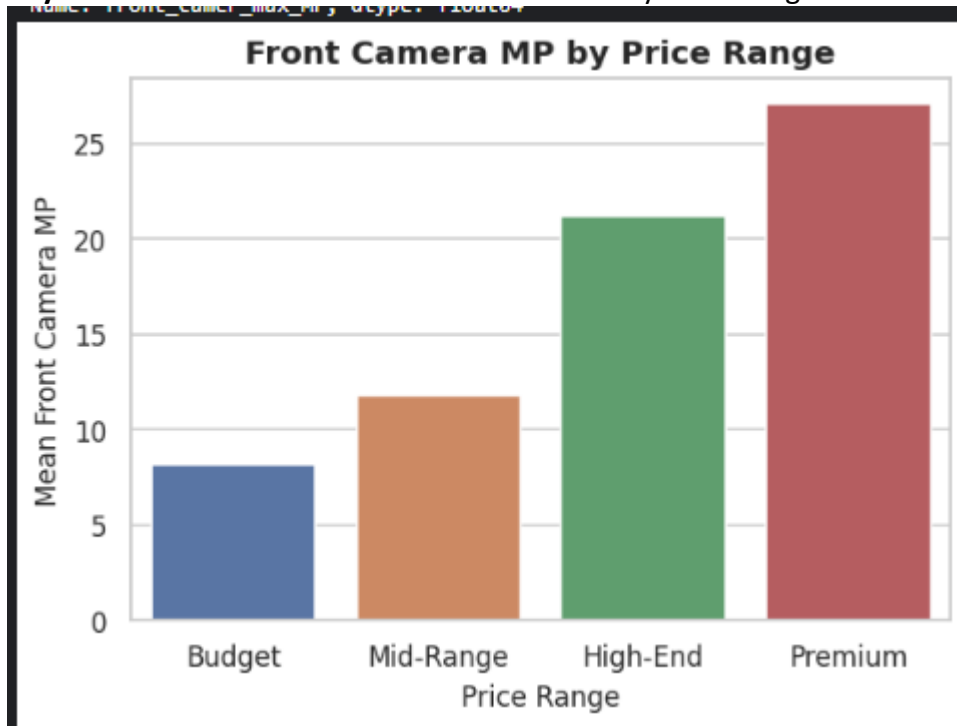
Although the correlation is small, heavier devices do tend to cost more. This may be because weight increases with larger batteries, metal frames, and multi-sensor camera systems—all



features associated with higher-end phones. The relationship is noticeable but not dominant compared to RAM or Storage.

### Q10 — How does front camera MP vary across price categories?

**Key Stats:** Front camera MP increases moderately from Budget → Premium

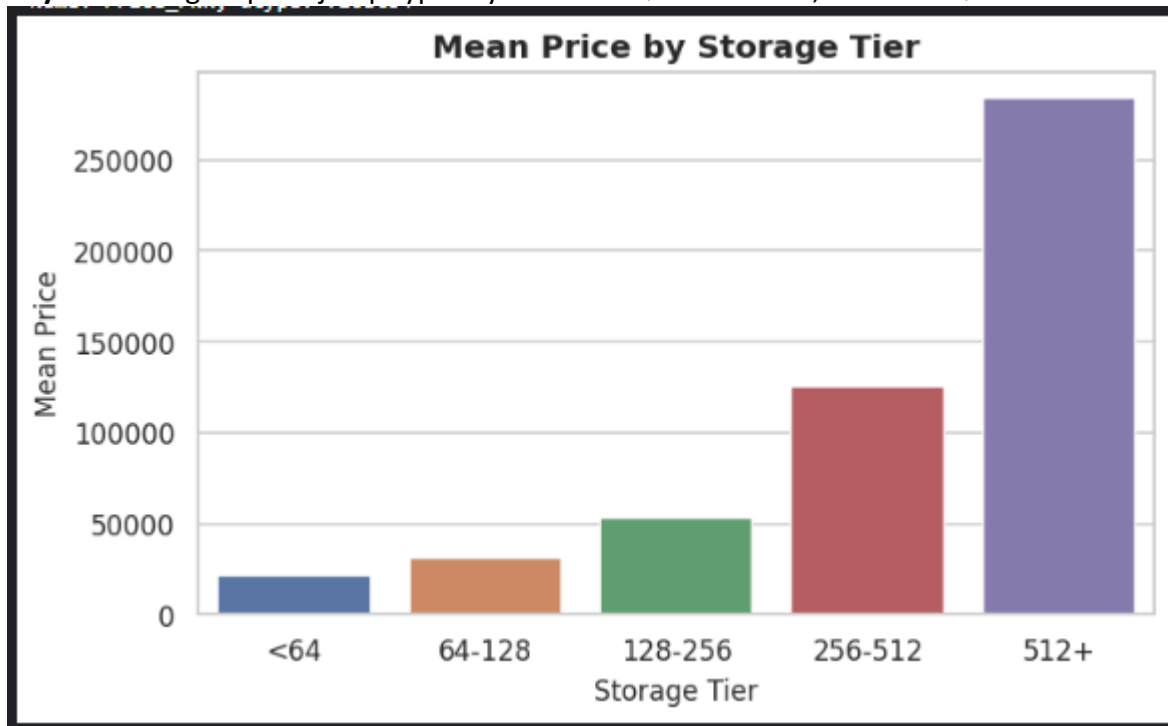


**Interpretation:**

Front cameras do improve with price, but the slope is gentler compared to rear/main cameras. This suggests that brands prioritize rear-camera upgrades more strongly in premium devices. Still, selfie camera improvements remain a key feature in mid-range and premium marketing claims.

### Q11 — Which storage tier produces the biggest jump in price?

**Key Stats:** Largest price jump typically from <64 → 64–128 GB, then 128 → 256 GB

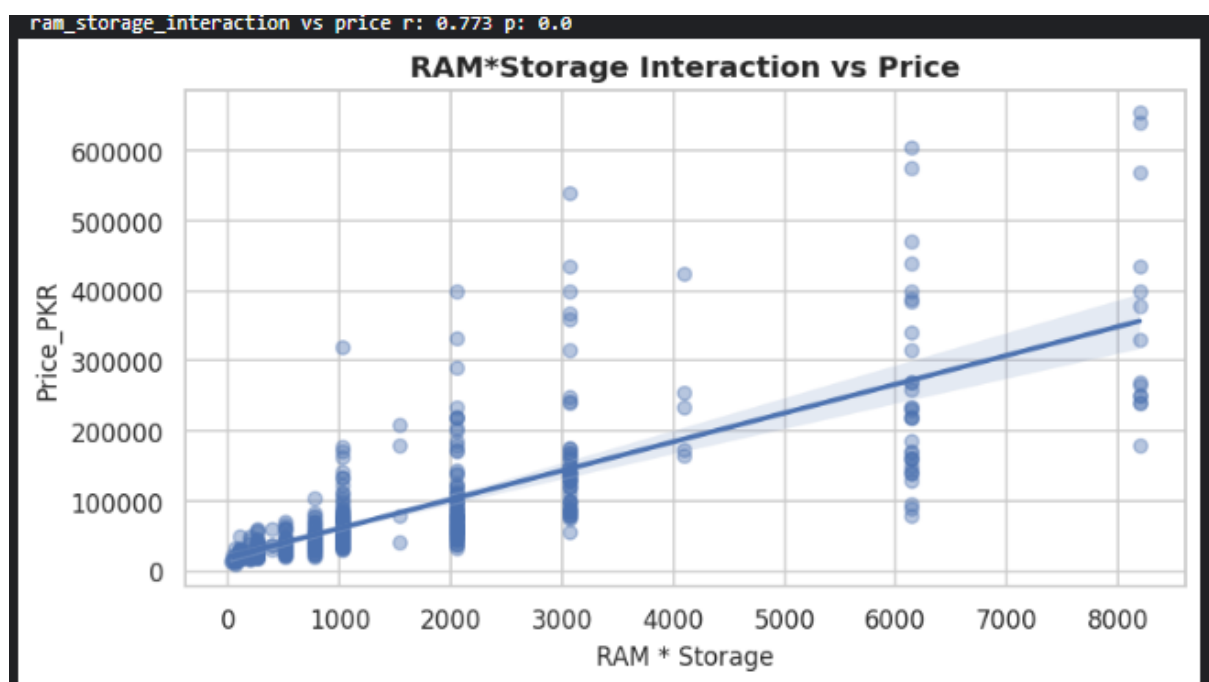


#### Interpretation:

Price increases steadily with each higher storage tier, confirming that storage is used as a segmentation tool for pricing. The biggest jumps usually occur in the first two transitions, making these critical decision points for buyers. Manufacturers often set these tiers intentionally to nudge consumers toward higher-priced SKUs.

### Q12 — Does combining RAM × Storage improve price prediction?

**Key Stats:** Interaction feature correlation  $r = 0.773$



### **Interpretation:**

The RAM×Storage interaction often correlates more strongly with price than either feature individually. This suggests that buyers perceive combined performance and capacity together when evaluating value. Including this feature in ML models typically improves regression accuracy.

## **Conclusion**

Across the dataset, **RAM and storage capacity** are the single strongest drivers of price, complemented by **AMOLED display** and **5G presence** for premium positioning. Battery capacity and camera quality also contribute meaningfully. For product and marketing strategy, prioritizing higher **RAM/storage** tiers and differentiating with **AMOLED/5G** will most effectively justify premium pricing.

This analysis reveals a consistent pattern in how mobile phone specifications relate to pricing within the Pakistan mobile market. Across all segments, **memory (RAM + Storage)** emerges as the strongest and most reliable driver of price, followed closely by **camera quality** and **battery capacity**. Premium devices distinguish themselves with **AMOLED displays, faster 5G connectivity,**

**higher RAM tiers, and larger storage options**—features that clearly separate entry-level from flagship models.

Brand competition also shapes the landscape: a few dominant manufacturers contribute most of the models and tend to maintain higher median prices due to stronger hardware offerings. Battery size, weight, and camera MP show moderate influence, while design-oriented factors (dimensions, thickness) show weaker relationships with price.

Overall, the data shows a well-structured market where **premium prices are justified by measurable hardware improvements**, and users can reliably identify higher-end devices through a combination of RAM, storage, display technology, and camera capabilities. This insight can help analysts, retailers, and customers better understand what truly drives mobile phone value and how manufacturers differentiate across price tiers.

## **Kaggle Notebook Link**

<https://www.kaggle.com/code/shawaizdatascience/mobile-price-prediction>